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$$\begin{aligned}f(x) &= \frac{2x^2 - 4x + 1}{\sqrt{x}} \\f'(x) &= \frac{g \cdot f' - f \cdot g'}{g^2} = \frac{\sqrt{x}(4x - 4) - (2x^2 - 4x + 1)\frac{1}{2}x^{-\frac{1}{2}}}{\sqrt{x}} \\&= \frac{2x(4x - 4) - (2x^2 - 4x + 1)}{2\sqrt{x}} \\&= \frac{8x^2 - 8x - 2x^2 + 4x - 1}{2x\sqrt{x}} \\&= \frac{6x^2 - 4x - 1}{2x\sqrt{x}}\end{aligned}$$

References